Example Presentation

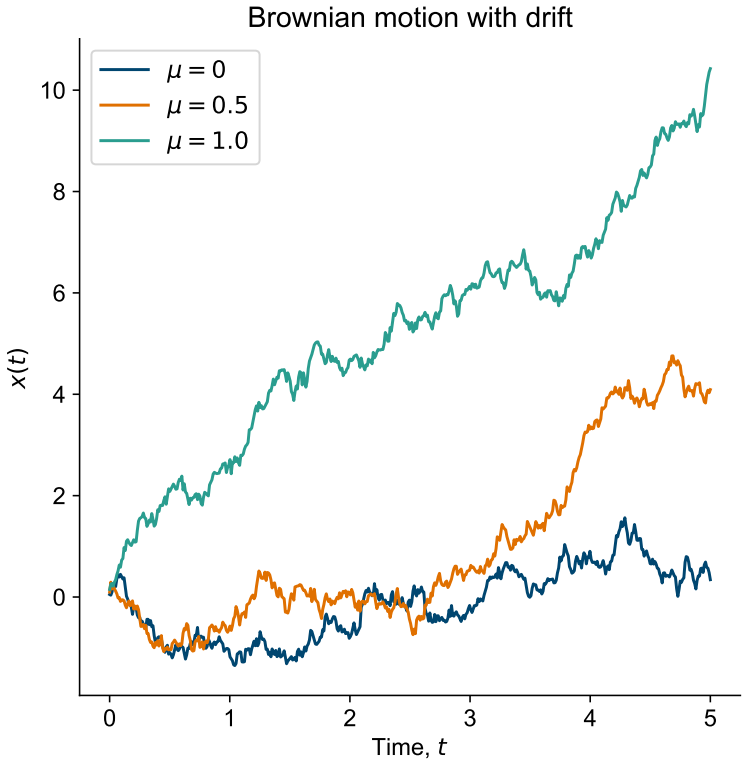
EE Presentations

Brownian motion with drift

A particle following Brownian motion with drift μ and volatility σ satisfies

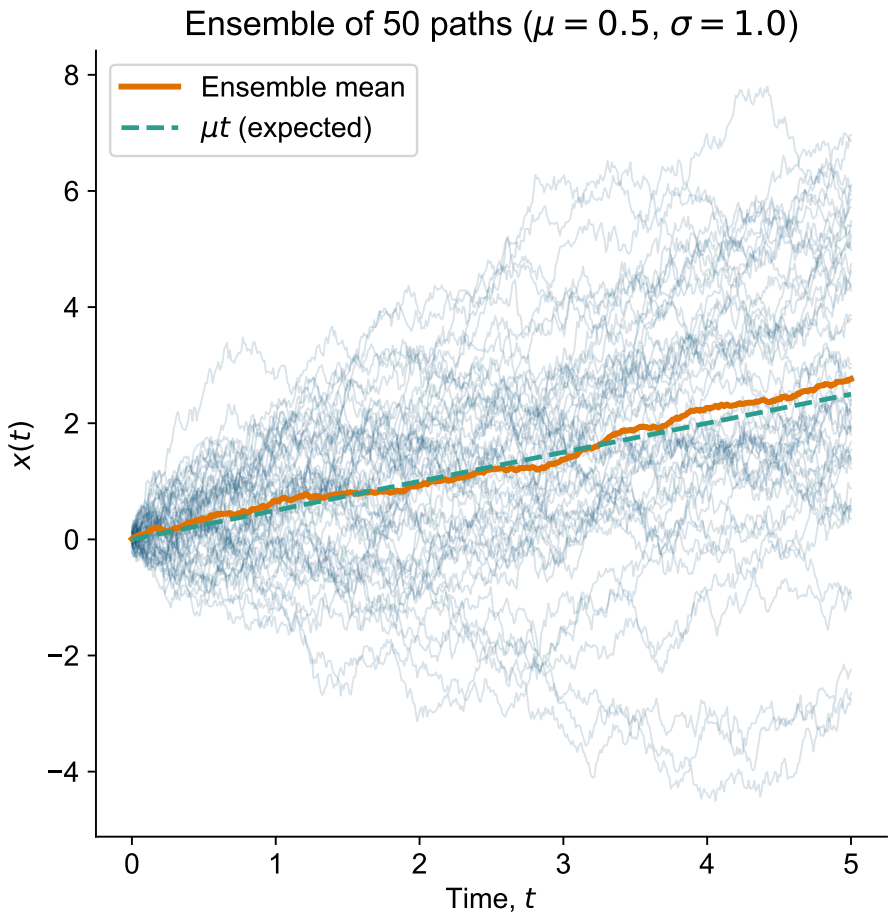
$$dx = \mu\, dt + \sigma\, dW\,.$$

The figure below shows three sample paths with different drift values.



Ensemble averages

For an ensemble of N independent paths, the sample mean converges to μt.



Key observations

- The ensemble average grows linearly: $\langle x(t) \rangle = \mu t$
- Individual paths fluctuate around this mean
- Higher σ increases the spread of paths
- The time average for a single path converges to μt only as $t \to \infty$

This distinction between ensemble and time averages is central to ergodicity economics.

Thank you.